POWER QUERY IN EXCEL

# Understanding Joins with Practical Tables

Merge Queries • Left • Right • Inner • Full Outer • Anti Joins

Session goal: combine two tables using a common key (SID) and interpret how each join changes the result.

Excel Data → Get Data / From Table/Range → Power Query Editor → Merge Queries

# Learning objectives

By the end of the session, learners should be able to:

1. **Identify the matching key** Use SID to connect student names with course records.
2. **Explain join types** Describe which rows are retained by each join.
3. **Perform Merge Queries** Choose tables, select matching columns, and expand results.
4. **Validate output** Check row counts, unmatched keys, and duplicate matches.

### Table 1 (Students)

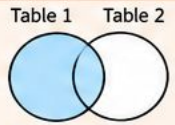
| SID | Name |
|---|---|
| 1 | Mayank |
| 2 | Raj |
| 3 | Nitin |
| null | Rajat |
| 4 | Animesh |

### Table 2 (Courses)

| SID | Course |
|---|---|
| 1 | A |
| 1 | B |
| 2 | A |
| 3 | C |
| null | A |
| null | B |
| 5 | C |
| 6 | B |

## 1. Left Join (Left Outer Join)

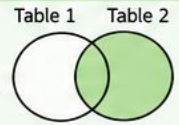
Returns all records from the left table and matching records from the right table.
If there is no match, right table columns are null.




| SID | Name | Course |
|---|---|---|
| 1 | Mayank | A |
| 1 | Mayank | B |
| 2 | Raj | A |
| 3 | Nitin | C |
| null | Rajat | A |
| null | Rajat | B |
| 4 | Animesh | null |

## 2. Right Join (Right Outer Join)

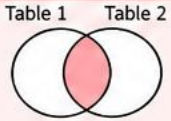
Returns all records from the right table and matching records from the left table.
If there is no match, left table columns are null.




| SID | Name | Course |
|---|---|---|
| 1 | Mayank | A |
| 1 | Mayank | B |
| 2 | Raj | A |
| 3 | Nitin | C |
| null | Rajat | A |
| null | Rajat | B |
| 5 | null | C |
| 6 | null | B |

## 3. Inner Join

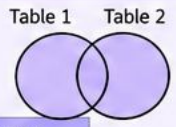
Returns only the matching records from both tables.




| SID | Name | Course |
|---|---|---|
| 1 | Mayank | A |
| 1 | Mayank | B |
| 2 | Raj | A |
| 3 | Nitin | C |
| null | Rajat | A |
| null | Rajat | B |

## 4. Full Outer Join

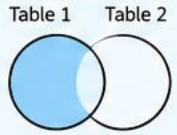
Returns all records when there is a match in either left or right table.
If there is no match, the missing side is null.




| SID | Name | Course |
|---|---|---|
| 1 | Mayank | A |
| 1 | Mayank | B |
| 2 | Raj | A |
| 3 | Nitin | C |
| null | Rajat | A |
| null | Rajat | B |
| 4 | Animesh | null |
| 5 | null | C |
| 6 | null | B |

## 5. Left Anti Join

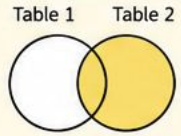
Returns records from the left table that do not have a match in the right table.




| SID | Name |
|---|---|
| 4 | Animesh |

## 6. Right Anti Join

Returns records from the right table that do not have a match in the left table.




| SID | Course |
|---|---|
| 5 | C |
| 6 | B |

# Why do we join tables?

Business data is often stored in separate tables.

| SID | Name |
|---|---|
| 1 | Mayank |
| 2 | Raj |
| 3 | Nitin |
| (blank) | Rohit |
| 4 | Ramesh |

**Table 1: student list**

**MERGE ON SID →**

| SID | Course |
|---|---|
| 1 | A |
| 1 | B |
| 2 | A |
| 3 | C |
| (blank) | A |
| (blank) | B |
| 5 | C |
| 6 | B |

**Table 2: course records**

**SID is the join key. Name and Course are descriptive columns.**

# Source Table 1 — Students

This is the left table in our examples.

| SID | Name |
|---|---|
| 1 | Mayank |
| 2 | Raj |
| 3 | Nitin |
| (blank) | Rohit |
| 4 | Ramesh |

## Observations

- SID 1–4 are numeric identifiers.
- One SID is blank/null.
- Names should be retained in a left join.

# Source Table 2 — Course records

This is the right table in our examples.

| SID | Course |
|---|---|
| 1 | A |
| 1 | B |
| 2 | A |
| 3 | C |
| (blank) | A |
| (blank) | B |
| 5 | C |
| 6 | B |

## Observations

- SID 1 appears twice (A and B).
- Some SIDs have no match in Table 1.
- Blank/null SID appears more than once.

# Merge Queries: the workflow

Merge matches rows across two tables; it does not simply stack them.

### 1 Load tables

Data → From Table/Range

### 2 Open Merge

Power Query Editor → Home → Merge Queries

### 3 Select key

Click SID in both table previews

### 4 Choose join

Select Left, Right, Inner, Full Outer, or Anti

### 5 Expand

Use expand icon to select Name/Course columns

### 6 Load

Close & Load to an Excel worksheet

# 1. Left Join (Left Outer)

Keep every row from Table 1; bring in matching records from Table 2.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| 1 | Mayank | 1 | A |
| 1 | Mayank | 1 | B |
| 2 | Raj | 2 | A |
| 3 | Nitin | 3 | C |
| (blank) | Rohit | (blank) | A |
| (blank) | Rohit | (blank) | B |
| 4 | Ramesh | (blank) | (blank) |

## Interpretation

- All student rows are retained.
- Unmatched students show blank Course.
- SID 1 appears twice because it has two course records.

# 2. Right Join (Right Outer)

Keep every row from Table 2; add names where SID matches Table 1.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| 1 | Mayank | 1 | A |
| 1 | Mayank | 1 | B |
| 2 | Raj | 2 | A |
| 3 | Nitin | 3 | C |
| (blank) | Rohit | (blank) | A |
| (blank) | Rohit | (blank) | B |
| (blank) | (blank) | 5 | C |
| (blank) | (blank) | 6 | B |

## Interpretation

- Every course record is retained.
- SIDs 5 and 6 have no student name.
- Repeated SID 1 creates multiple output rows.

# 3. Inner Join

Keep only rows where SID exists in both tables.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| 1 | Mayank | 1 | A |
| 1 | Mayank | 1 | B |
| 2 | Raj | 2 | A |
| 3 | Nitin | 3 | C |
| (blank) | Rohit | (blank) | A |
| (blank) | Rohit | (blank) | B |

## Interpretation

- Unmatched SIDs are removed.
- SID 1 returns two rows (A and B).
- Use when only matched records are required.

# 4. Full Outer Join

Keep all rows from both tables, whether matched or not.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| 1 | Mayank | 1 | A |
| 1 | Mayank | 1 | B |
| 2 | Raj | 2 | A |
| 3 | Nitin | 3 | C |
| (blank) | Rohit | (blank) | A |
| (blank) | Rohit | (blank) | B |
| 4 | Ramesh | (blank) | (blank) |
| (blank) | (blank) | 5 | C |
| (blank) | (blank) | 6 | B |

## Interpretation

- Matched records show Name and Course.
- Unmatched rows have blank columns.
- Useful for reconciling two lists.

# 5. Left Anti Join

Return only Table 1 rows with no matching SID in Table 2.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| 4 | Ramesh | (blank) | (blank) |

## How to interpret

These are students whose SID has no matching course record. Anti joins are useful for exception reports and data-quality checks.

# 6. Right Anti Join

Return only Table 2 rows with no matching SID in Table 1.

| Table1.SID | Table1.Name | Table2.SID | Table2.Course |
|---|---|---|---|
| (blank) | (blank) | 5 | C |
| (blank) | (blank) | 6 | B |

## How to interpret

These are course records whose SID has no matching student record. Use this to find orphan records that need investigation.

# Join types at a glance

Choose the join according to the question you want to answer.

| Join type | Rows retained | Example business question |
|---|---|---|
| Left Outer | All Table 1 rows | Find every student and any available course |
| Right Outer | All Table 2 rows | Keep every course record |
| Inner | Only matches | Find records present in both tables |
| Full Outer | All rows from both | Reconcile two lists |
| Left Anti | Unmatched Table 1 | Find students with no course record |
| Right Anti | Unmatched Table 2 | Find courses with no student record |

# Why do some rows appear more than once?

A join can produce multiple rows when a key has multiple matches.

| SID | Name |
|---|---|
| 1 | Mayank |

1

↓

2 output rows

| SID | Course |
|---|---|
| 1 | A |
| 1 | B |

**If one key matches multiple rows, the output repeats the left-side record for each match. Check whether that is expected.**

# Important: blank / null keys

Blank keys need special attention before merging.

**In this teaching workbook, blank SID values are treated as matching to mirror the supplied example.**

## In a real dataset:

- Inspect null keys in both tables before merging.
- Do not assume a blank SID identifies the same person or record.
- Consider filtering null keys or replacing them with a controlled exception label.
- Validate output counts and investigate unexpected matches.

# Classroom practical — follow along in Excel

Use Power_Query_Joins_Practice.xlsx and compare each result sheet.

1. Load Source_Table1 and Source_Table2 into Power Query.
2. Merge on SID using Left Outer and expand Course.
3. Repeat using Inner and Full Outer joins.
4. Use Left Anti and Right Anti to identify unmatched records.
5. Compare your output row counts with the example worksheets.
6. Close & Load the final result to Excel.

# Validation checklist after a merge

A correct-looking table is not enough—check the output.

| Check | What to verify |
| --- | --- |
| Key columns | Confirm SID is selected in both queries. |
| Data types | Both SID columns should have compatible data types. |
| Row count | Explain increases/decreases using the join type and duplicate keys. |
| Unmatched records | Use Anti joins or nulls in expanded columns to investigate. |
| Duplicate keys | Check if duplicate SID values are expected. |
| Refresh | Use Refresh All after source data changes. |

# Quick knowledge check

Ask learners to answer before revealing the next slide.

**1. Which join keeps all rows from Table 1?**

2. Which join keeps only matched SIDs?

**3. Which join helps find students with no course record?**

4. Why does SID 1 appear twice after a merge?

**5. What is the difference between Merge and Append?**

# Knowledge check — answers

Use these to recap the core concepts.

| Question | Answer |
|---|---|
| 1 | Left Outer Join |
| 2 | Inner Join |
| 3 | Left Anti Join |
| 4 | Because SID 1 has two matching course rows (A and B). |
| 5 | Merge adds columns by matching keys; Append stacks rows. |

KEY TAKEAWAYS

# Joins are about which rows to keep.

Choose the join based on the question, then validate duplicates, null keys, and row counts.

Practice the same joins in Excel Power Query and compare with the provided workbook.

Thank you